Global Positioning System (GPS) – General Data

1. General

The Global Positioning System (GPS) consists of one or two independent KGS 200 Global Navigation System Sensor Unit (GNSSU) receivers, and one or two GPS antennas. GPS 1 and GPS 2 (if installed), collect and compile signals from satellites and provide the aircraft avionics systems with position, velocity, and time. GPS 1 is part of the basic aircraft avionics suite. Provisions for GPS 2 have been made, and can be installed as part of the IFR Operations Bundle customer option order 6209.

Mod 6/2304 upgrades the KGS 200 GPS receivers to KGS 200 SBAS GPS receivers.

For aircraft with S.O.O. 6388 incorporated, there is one Post Mod 6/2304 KGS 200 Global Navigation System Sensor Unit (GNSSU) receiver installed.

For aircraft with S.O.O. 6388 incorporated, a second Post Mod 6/2304 KGS 200 Global Navigation System Sensor Unit (GNSSU) and associated systems are installed on the aircraft by incorporating customer Standard Order Option S.O.O. 6380.

2. Description

A. KGS 200 GNSSU System

1) Global Positioning System (KGS 200) (Mod 6/2049) (Pre Mod 6/2304)

One or two independent KGS 200 Global Navigation System Sensor Unit (GNSSU) receivers are installed in the nose avionics bay. The receiver uses 24 channels grouped in 3 hardware groups each consisting of 8 channels. One group of 8 channels is used for navigation at any given time. The GPS sensor unit calculates and outputs navigation data, satellite measurement data, Receiver Autonomous Integrity Monitoring (RAIM) and Predictive RAIM (PRAIM). The processed output data is sent to the CSIO module in the MAU via ARINC 429 for use by the rest of the avionics systems. The sensor data unit also manages Sign Status Matrix (SSM), satellite status and does Built-In Test Equipment (BITE) functions. An Apex maintenance function interfaces with the GPS sensor. The GPS data page can be accessed from the Sensors page. The Sensors page can be accessed with the systems MFD lower left window in focus and selecting the Sensors page menu. The No. 1 KGS 200 GNSSU provides GPS position and GPS altitude to the EGPWS.

28 Volt DC power is protected by a 3 Amp GPS 1 (J3) (Left Bus) and GPS 2 (J4) (Right Bus) (if installed) circuit breaker located on the avionics circuit breaker panel (refer to 24-60-00, DC Electrical Load Distribution – General Data).

There are one or two GPS antennas mounted on the top of the fuselage. GPS 1 antenna is mounted just off the centerline to the starboard side at FS140 and GPS 2 antenna, if installed, is mounted off the centerline to the port side at FS315. The GPS antenna is a low profile, tear drop type and has a single coaxial connector.

The KGS 200 GNSSU interfaces to the following aircraft systems:

a) Modular Avionics Unit (MAU) (refer to 31-43-00, Modular Avionics Unit (MAU) (Mod 6/2049) – General Data).

1 SmartView Synthetic Vision System (SVS) (S.O.O. 6222) (refer to 31-10-00, Instrument and Control Panels – General Data).

2 Flight Management System (FMS) (Mod 6/2303) (refer to 34-60-00, Flight Management System (FMS) – General Data).

b) Single or Dual (if installed) Transponder System (refer to 34-53-00, KXP 2290 Transponder System – General Data).

c) Enhanced Ground Proximity Warning System (TAWS) (Mod 6/2049) (refer to 34-44-00, Enhanced Ground Proximity Warning System (TAWS) – General Data), if installed.

d) Air Data, Attitude and Heading Reference System (ADAHRS) (KSG 7200) (refer to 34-23-00, Air Data, Attitude and Heading Reference System (ADAHRS) (KSG 7200) – General Data).

2) Global Positioning System (KGS 200) with SBAS Capability (Post Mod 6/2304)

Various GNSS systems around the world use a correction signal generated by ground sensors, transmitted via geosynchronous satellites to SBAS capable GPS receivers. This facilitates the conduct of more accurate GPS-based navigation and approaches.

Accurate aircraft positioning is aided by the Wide Area Augmentation System (WAAS) that was developed to augment GPS. WAAS utilizes ground-based reference stations and geostationary WAAS satellites in a system known as the Satellite Based Augmentation System (SBAS). WAAS is capable of achieving comparable Category 1 Instrument Landing System (ILS) precision approach accuracy.

Aircraft incorporating Mod 6/2304 have the KGS 200 GPS receivers with Satellite-based Augmentation System (SBAS) capability. The KGS 200 SBAS GPS receivers are direct replacements for the non-SBAS KGS 200 GPS receiver LRU's installed on the forward avionics shelf. No changes are required to the KA 96 GPS antennas or coaxial cables. GPS 1 and GPS 2 (if installed) circuit breaker size and location are not affected.

Installation of Mod 6/2304 (SBAS GPS Receivers) is a prerequisite for S.O.O. 6225, SBAS navigation and approach capability, as well as S.O.O. 6347, ADS-B Out functionality, refer to 34-53-00, KXP 2290 Transponder System – General Data.

The SBAS navigation and approach capability is enabled through the incorporation of customer standard order option S.O.O. 6225. When two GPS receivers are installed, both must be SBAS receivers. SBAS GPS approach capability allows SBAS-based GPS approaches such as LPV approaches.

For aircraft with S.O.O. 6388 incorporated, there is one Post Mod 6/2304 KGS 200 Global Navigation System Sensor Unit (GNSSU) receiver installed.

For aircraft with S.O.O. 6388 incorporated, a second Post Mod 6/2304 KGS 200 Global Navigation System Sensor Unit (GNSSU) and associated systems are installed on the aircraft by incorporating customer Standard Order Option S.O.O. 6380.

3. Adjustment/Test

A. Function Test Global Positioning System

1) General

These procedures are for the function test of global positioning systems GPS 1 and GPS 2 (if installed).

2) Function Test

a) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

b) Wait several minutes for the aircraft electrical and avionics systems to initialize.

c) In the Apex system Sensors page, verify GPS 1 and GPS 2 (if installed) receive data is valid.

d) Ensure there are no GPS Fail CAS messages or PFD annunciations.

e) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

B. ADS-B GPS Input Verification (Mod 6/2354 KXP 2290A with S.O.O. 6347 ADS-B Out)

1) General

These procedures are for the ADS-B Out GPS input verification test procedures for GPS 1 and GPS 2 (if installed).

2) ADS-B GPS Input Verification

a) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

b) Wait several minutes for the aircraft electrical and avionics systems to initialize.

c) Verify ADS-B status indication is white in the Transponder Radio Tuning window.

d) Open GPS 1 (J3) (Left Bus) circuit breaker, wait a few seconds for the GPS 1 Fail CAS message to be displayed. Ensure ADS-B status indication remains white, leave GPS 1 circuit breaker open for the next step (dual GPS). Ensure ADS-B status indication turns amber, skip the next step (single GPS).

e) Open GPS 2 (J4) (Right Bus) circuit breaker, wait a few seconds for the GPS 2 Fail CAS message to be displayed, ensure ADS-B status indication turns amber.

f) Close GPS 1 (J3) (Left Bus) circuit breaker, allow enough time for GPS 1 to initialize, ensure ADS-B status indicator returns to white.

g) Close GPS 2 (J4) (Right Bus) circuit breaker (dual GPS).

h) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

C. Electrical Bonding Test – GPS 1 and GPS 2 (if installed) Antenna

1) General

These procedures are for the electrical bonding test procedures for GPS 1 antenna and GPS 2 antenna (if installed).

2) Electrical Bonding Test

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Gain access to the ceiling compartment forward of station 143.95 for GPS 1 antenna and forward of station 321.6 for GPS 2 antenna (if installed). (Refer to 25-20-00, Cabin Compartment – Commuter Version (Mod 6/2050) – General Data).

c) Verify that the electrical conductivity is consistent over the electrical bond contact area by placing a probe on the GPS antenna mounting screw. Place the second probe on the nearest ground stud. The resistance between the two points must be < 3.0 milliohms unless stated otherwise.

d) If the electrical conductivity does not meet requirements, check the antenna electrical bonding in accordance with 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data.

e) Reinstall the ceiling compartment interior. (Refer to 25-20-00, Cabin Compartment – Commuter Version (Mod 6/2050) – General Data).

4. Removal/Installation

WARNING **INSTALLATION OF THIS COMPONENT CONTAINS PROTECTION FEATURES THAT ARE CRITICAL TO PREVENTING DEGRADATION FROM THE EFFECTS OF HIGH INTENSITY RADIATED FIELDS (HIRF) AND INDIRECT EFFECTS OF LIGHTNING (IEL). ENSURE INTEGRITY OF THE HIRF/IEL PROTECTION FEATURES ARE MAINTAINED IN ACCORDANCE WITH THE FOLLOWING:**

CHASSIS GROUNDING – ALL RACK MOUNT / REMOTE MOUNT UNITS SHALL BE ELECTRICALLY BONDED TO THE AIRFRAME. ENSURE THAT THE MATING SURFACES BETWEEN THE LRU MOUNTING TRAY (OR LRU MOUNTING FEET IF A TRAY IS NOT USED) PROVIDE A LOW IMPEDANCE ELECTRICAL PATH. ALL MATING SURFACES MUST BE FREE OF ALL PAINT AND OTHER NONCONDUCTIVE ELEMENTS. REFER TO BONDING REQUIREMENTS IN 20-75-00, ELECTRICAL BONDING AND GROUNDING OF AIRCRAFT STRUCTURES – GENERAL DATA.

SHIELD GROUNDS – VERIFY SHIELD GROUNDING AT CONNECTOR BACKSHELLS AS PER REQUIREMENTS IN PSM 1-64-2W.

BACKSHELLS OR STRAIN RELIEFS – VERIFY PROPER ELECTRICAL BONDING BETWEEN THE BACKSHELL/STRAIN RELIEF AND THE MOUNTING TRAY OR LRU CASE AS PER REQUIREMENTS IN PSM 1–64–2W.

A. Removal/Installation of GPS Receiver (KGS 200 GNSSU)

1) General

These procedures are for the removal and installation of the GPS 1 receiver and GPS 2 receiver (if installed).

2) Removal

a) Ensure the aircraft power is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Pull the GPS 1 (J3) (Left Bus) and/or GPS 2 (J4) (Right Bus), circuit breakers on the avionics circuit breaker panel and fit safety clips.

c) Gain access to the forward avionics shelves through the forward baggage compartment.

d) Remove top, middle and bottom covers attached to bulkhead station 44.0 for full access.

e) Disconnect the electrical and coaxial cables at the port side rear of the GPS receiver unit.

f) Remove the four screws that hold the GPS receiver unit to the bottom of the middle avionics shelf.

g) Remove unit from aircraft.

Note Stow the connectors of the removed components.

3) Installation

a) Ensure the aircraft power is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure that GPS 1 (J3) (Left Bus) and/or GPS 2 (J4) (Right Bus), circuit breakers on the avionics circuit breaker panel are open and fitted with safety clips.

c) Align the GPS receiver mounting holes to the bottom of the middle avionics shelf. Ensure correct orientation, connectors facing inward.

d) Install the four mounting screws and ensure receiver unit is securely held.

e) Reconnect the electrical and coaxial cables at the port side rear of the GPS receiver unit. Ensure connections are sound and secure.

f) Reinstall top, middle and bottom covers attached to bulkhead station 44.0 at the aft wall of the baggage compartment. Ensure covers are mounted correctly and secure.

WARNING **ACCESS COVERS ARE REQUIRED TO MAINTAIN PROTECTION FEATURES THAT ARE ESSENTIAL TO MINIMIZING ADVERSE EFFECTS FROM HIGH INTENSITY RADIATED FIELDS (HIRF).**

g) Remove safety clips and reset circuit breakers GPS 1 (J3) (Left Bus) and/or GPS 2 (J4) (Right Bus), on the avionics circuit breaker panel.

h) Do a function test on the Global Positioning System, refer to Para 3.A.

i) Do an ADS-B GPS Input Verification, refer to Para 3.B (applicable for Mod 6/2354 with S.O.O. 6347).

B. Removal/Installation of GPS Antenna (KA96)

1) General

These procedures are for the removal and installation of the GPS 1 and GPS 2 (if installed) antennas.

2) Removal

a) Ensure the aircraft power is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Pull the GPS 1 (J3) (Left Bus) and/or GPS 2 (J4) (Right Bus), circuit breakers on the avionics circuit breaker panel and fit safety clips.

c) Carefully cut the sealant from around the base of the antenna.

d) Support the antenna and remove the four screws from the base of the antenna.

e) Carefully lift the antenna clear of the fuselage until there is access to the coaxial connection.

f) Disconnect the coaxial connector.

g) Remove antenna from aircraft.

Note Stow connectors of the removed components.

3) Installation

a) Ensure the aircraft power is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure that GPS 1 (J3) (Left Bus) and/or GPS 2 (J4) (Right Bus), circuit breaker on the avionics circuit breaker panel are open and fitted with safety clips.

c) Solvent clean the electrical contact bond areas immediately before the installation (refer to 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data).

d) Bring the antenna assembly close to the fuselage, connect the coaxial connector.

e) Secure the antenna assembly on fuselage with the four mounting screws.

f) Solvent clean the exposed area of the airframe around the antenna (refer to 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data).

g) Touch up the exposed area of airframe around the antenna if required with F19 primers by brush within 24 hours of the antenna installation. Ensure that the primer re-finishing provides complete coverage of the affected area.

h) If the affected area requires paint touch up, brush coat the area with the top-coat paint system matching the existing.

i) Use sealant to both fillet seal around the antenna base and dome seal the securing screws. The aircraft will be ready to fly approximately 24 hours after sealing (refer to 20-80-00, Installation of Aircraft Antennas – General data).

j) Check the antenna electrical bonding, refer to Para 3.C.

k) Remove safety clips and reset circuit breakers GPS 1 (J3) (Left Bus) and/or GPS 2 (J4) (Right Bus) on the avionics circuit breaker panel.

l) Do a function test on the Global Positioning System, refer to Para 3.A.

5. Troubleshooting

A. Global Positioning System (KGS 200 GNSSU)

If the GPS is deemed inoperative check the following:

1) Ensure GPS 1 (J3) and/or GPS 2 (J4) circuit breaker are closed.

2) Visually inspect GPS connector for damaged wires or contacts.

3) Visually inspect GPS antenna for corrosion and contaminants.

4) Ensure that the antenna coaxial cable connector is properly mated and secure.

5) If troubleshooting at the wire level is required refer to the applicable wiring diagrams in Wiring Manual PSM 1-64-2W.

6) If repair or service is required, send to appropriate rated avionics repair facility.